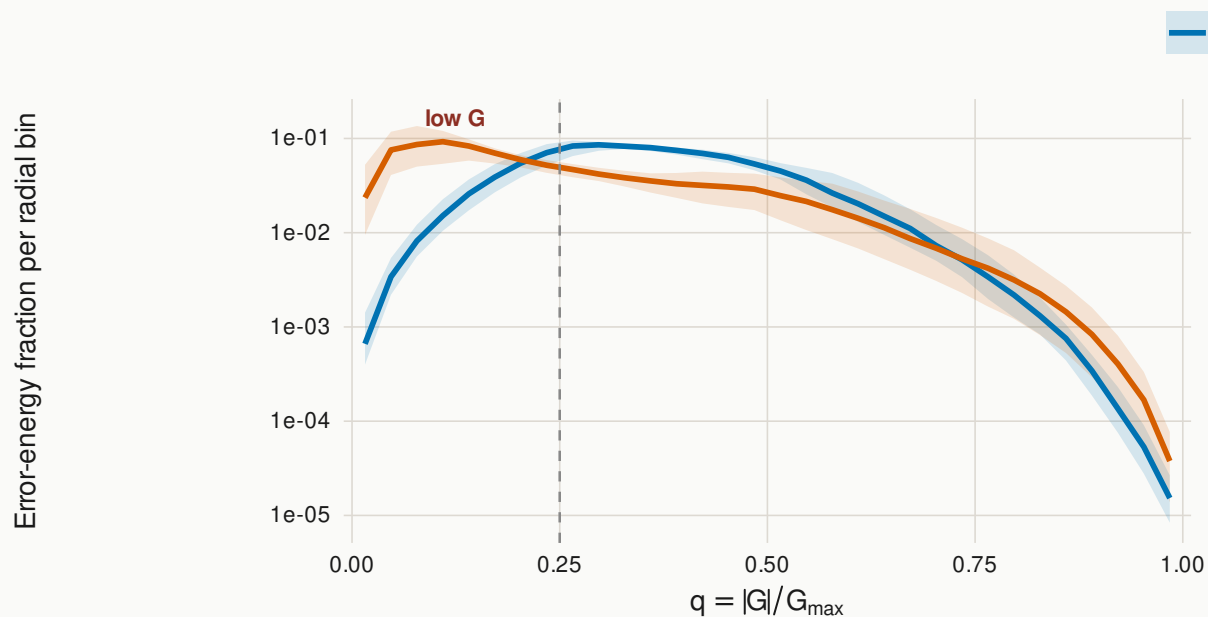


Figure 7 | Frequency allocation of reconstruction error controls Hartree fidelity at matched L_∞

Exact ZFP/SZ3 common-support audit: 457 matched pairs across 214 materials. The Nyquist-safe operator reproduces the historical codec effect while restoring exact Fourier/real-space parity.

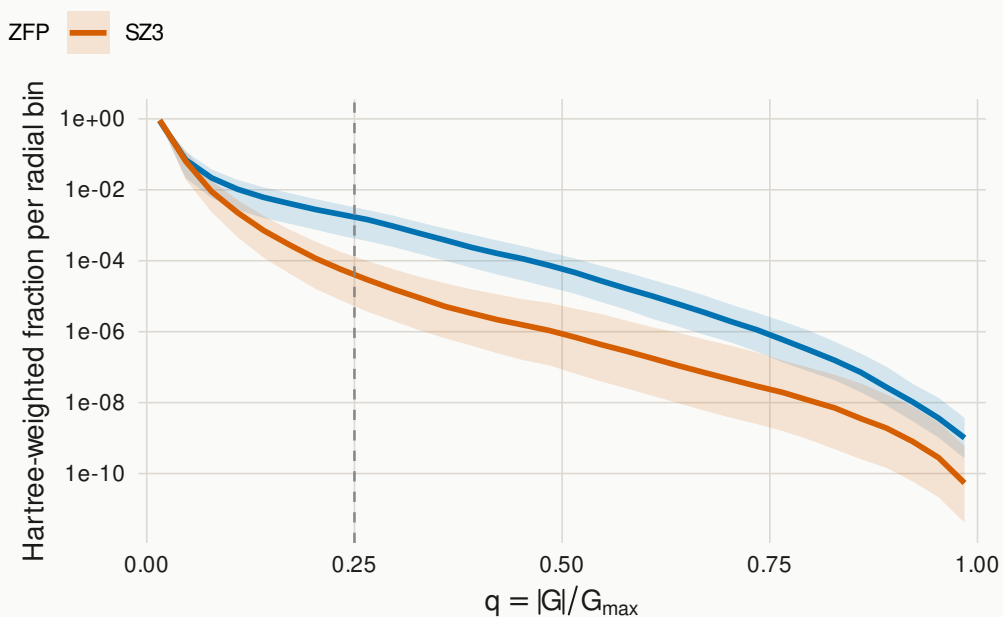
A | ZFP shifts reconstruction error away from low G

Median radial error-energy fraction; ribbons show pairwise IQR across 457 matched pairs



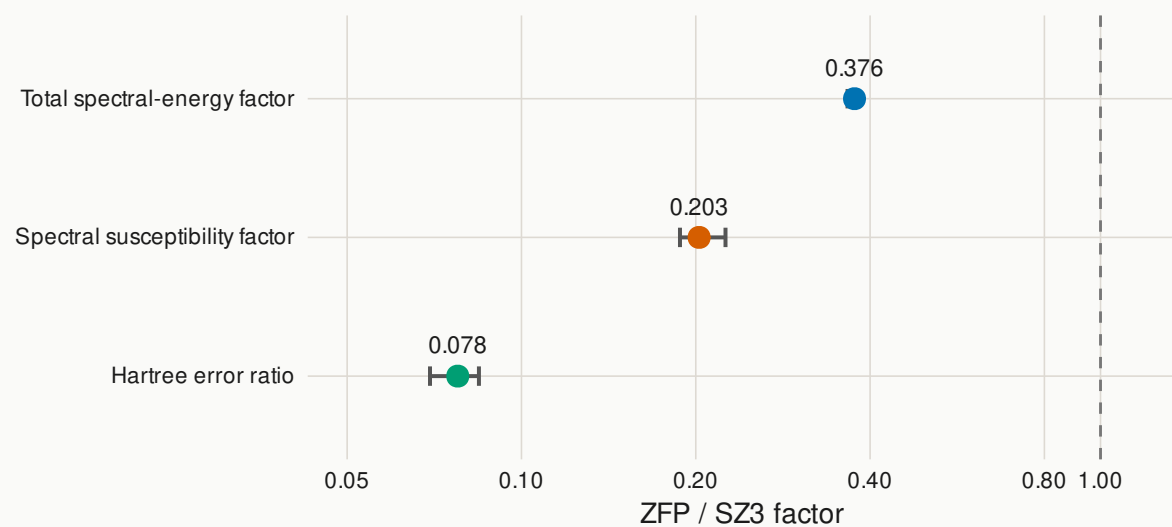
B | $|G|^{-4}$ weighting makes low-G error decisive

Median fraction of Hartree-weighted spectral energy in each radial bin



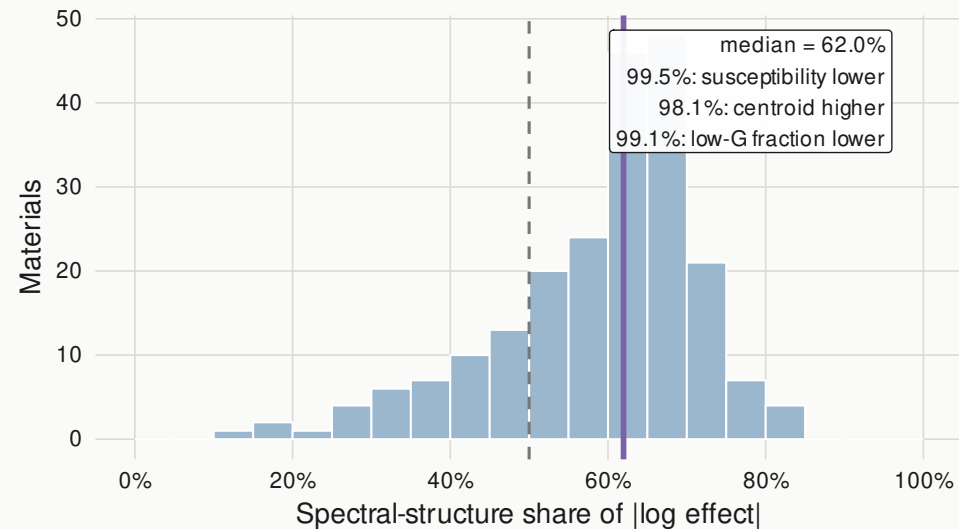
C | The matched-distortion effect separates into magnitude and structure

Material-level centers with 95% bootstrap intervals; exact multiplicative closure is pairwise



D | Frequency structure dominates across materials

Within-material median absolute-log share attributed to spectral Hartree susceptibility



Historical Hartree ratio = 0.0776221; Nyquist-safe ratio = 0.0776219; maximum Parseval relative error = 1.30×10^{-15} . For each matched pair, $R_H = \sqrt{E_{\text{ZFP}}/E_{\text{SZ3}}} \times \sqrt{S_{H,\text{ZFP}}/S_{H,\text{SZ3}}}$, with $S_H = [\sum |\Delta \rho(G)|^2 / |G|^4] / [\sum |\Delta \rho(G)|^2 / |G|^4]$.